PERFECT OBJECTS OVER LOCALLY RINGED SPACES ARE FINITE COMPLEXES OF FINITE LOCALLY FREE MODULES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Representatives and perfection. Cohomology of Sheaves, Lemma cohomology-lemma-perfect-independent-representative. Perfection from a covering by strictly perfect representatives.
- Summands of finite free are locally free. Modules, Lemma modules-lemma-direct-summand-of-locally-free-is-locally-free. On locally ringed spaces.

Source reference. Original lemma 08CN: sources/stacks-chow/tags.txt, source lines 10608–10608. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BCI: sources/stacks-chow/tags.txt, source lines 14281–14281. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let E be an object of $D(\mathcal{O}_X)$. Assume that all stalks $\mathcal{O}_{X,x}$ are local rings. Then the following are equivalent*

(1) *E is perfect,*
(2) *there exists an open covering $X = \bigcup U_i$ such that $E|_{U_i}$ can be represented by a finite complex of finite locally free $\mathcal{O}_{U_i}$-modules, and*
(3) *there exists an open covering $X = \bigcup U_i$ such that $E|_{U_i}$ can be represented by a finite complex of finite free $\mathcal{O}_{U_i}$-modules.*

Proof. This follows from Lemma 08CN and the fact that on X every direct summand of a finite free module is finite locally free. See Modules, Lemma 0BCI. □

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